

REPO: <https://github.com/ttrimalo/CSE-464-2026-ttrimalo>

1. BFS:

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" ...  
==== BFS TEST ====  
Visit Node History: a  
Visit Node History: a-b  
Visit Node History: a-e  
Visit Node History: a-b-c  
Found target node: c
```

Main.java:

```
package org;
```

```
public class Main {  
    public static void main(String[] args) throws Exception {  
        GraphCommands g = new GraphCommands();  
        g.parseGraph("input.dot");  
        System.out.println("==== BFS TEST =====");  
        g.graphSearch("a", "c", Algorithm.BFS);  
    }  
}
```

2. DFS:

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" ...  
==== DFS TEST ====  
Visit Node History: a  
Visit Node History: a-e  
Visit Node History: a-e-g  
Visit Node History: a-e-g-h  
Visit Node History: a-e-f  
Visit Node History: a-b  
Visit Node History: a-b-c  
Found target node: c
```

Main.java:

```
package org;
```

```
public class Main {  
    public static void main(String[] args) throws Exception {  
        GraphCommands g = new GraphCommands();  
        g.parseGraph("input.dot");  
        System.out.println("==== DFS TEST =====");  
        g.graphSearch("a", "c", Algorithm.DFS);  
    }  
}
```

3. RandomWalk:

```
===== RANDOM WALK =====

Run 1
Visit Node History: a
Visit Node History: a-e
Visit Node History: a-e-g
Visit Node History: a-e-g-h
Reached dead end at node h

Run 2
Visit Node History: a
Visit Node History: a-b
Visit Node History: a-b-c
Found target node: c
```

Main.java:

```
package org;
```

```
public class Main {
    public static void main(String[] args) throws Exception {
        GraphCommands g = new GraphCommands();
        g.parseGraph("input.dot");
        System.out.println("===== RANDOM WALK =====");
        for(int i=1;i<=5;i++){
            System.out.println("\nRun " + i);
            g.graphSearch("a", "c", Algorithm.RANDOM);
        }
    }
}
```

4. GitHub Actions:

✓ **Course Project Part 3**

Jave CI #15: Pull request [#1](#) opened by [ttrimalo](#)

✓ **Added RandomWalkSearch**

Jave CI #14: Commit [fcdf86f](#) pushed by [ttrimalo](#)

✓ **Applied strategy pattern for BFS and DFS**

Jave CI #13: Commit [1312528](#) pushed by [ttrimalo](#)

✓ **Refactor 5: Create help function for choosing algorithm**

Jave CI #12: Commit [6ec93a8](#) pushed by [ttrimalo](#)

✓ **Refactor 4: Change LinkedList to ArrayDeque**

Jave CI #11: Commit [6bbf949](#) pushed by [ttrimalo](#)

✓ **Refactor 3: Use automatic closing for file writing**

Jave CI #10: Commit [ad9ec04](#) pushed by [ttrimalo](#)

✓ **Refactor 2: ensureVertexExists helper method**

Jave CI #9: Commit [f74aec9](#) pushed by [ttrimalo](#)

✓ **Refactor 1: Change graphCommands to GraphCommands**

Jave CI #8: Commit [d920143](#) pushed by [ttrimalo](#)

5. Pull Request:



ttrimalo commented [1 hour ago](#)Owner ...

5 refactoring commits
Template Method pattern for BFS/DFS
Strategy pattern for algorithm selection
Random Walk search algorithm





Tyler Trimaloff added 7 commits [2 hours ago](#)

[Refactor 1: Change graphCommands to GraphCommands](#)✓ [d920143](#)

[Refactor 2: ensureVertexExists helper method](#)✓ [f74aec9](#)

[Refactor 3: Use automatic closing for file writing](#)✓ [ad9ec04](#)

[Refactor 4: Change LinkedList to ArrayDeque](#)✓ [6bbf949](#)

[Refactor 5: Create help function for choosing algorithm](#)✓ [6ec93a8](#)

[Applied strategy pattern for BFS and DFS](#)✓ [1312528](#)

[Added RandomWalkSearch](#)✓ [fcd86f](#)



**ttrimalo** started a reviewPending View reviewed changes

▼ `src/main/java/org/GraphCommands.java`

13

- import java.util.*;

14

13

15

- public class graphCommands{

14

+ public class GraphCommands{



ttrimalo PendingOwner Author ...

Changes to GraphCommands for grammatically correct java code





ttrimalo PendingOwner Author ...

Originally had it as graphCommands, so capitalizing the G makes it a grammatically correct Java class.

6. Commits on refactor branch:

 **refactor** ▾

 Commits on Apr 27, 2026

Added RandomWalkSearch
 Tyler Trimaloff committed 1 hour ago ·  2 / 2

Applied strategy pattern for BFS and DFS
 Tyler Trimaloff committed 1 hour ago ·  1 / 1

Refactor 5: Create help function for choosing algorithm
 Tyler Trimaloff committed 1 hour ago ·  1 / 1

Refactor 4: Change LinkedList to ArrayDeque
 Tyler Trimaloff committed 1 hour ago ·  1 / 1

Refactor 3: Use automatic closing for file writing
 Tyler Trimaloff committed 1 hour ago ·  1 / 1

Refactor 2: ensureVertexExists helper method
 Tyler Trimaloff committed 2 hours ago ·  1 / 1

Refactor 1: Change graphCommands to GraphCommands
 Tyler Trimaloff committed 2 hours ago ·  1 / 1

Commits:

Refactor 1:

<https://github.com/ttrimalo/CSE-464-2026-ttrimalo/commit/d9201436bd5023893ab00e5398cd9250318f4ab9>

Refactor 2:

<https://github.com/ttrimalo/CSE-464-2026-ttrimalo/commit/f74aec9a0c9f3722d5264382852cb1aebd0de8bf>

Refactor 3:

<https://github.com/ttrimalo/CSE-464-2026-ttrimalo/commit/ad9ec04e9e9a96555b1cc3084ee337ee15cbd83e>

Refactor 4:

<https://github.com/ttrimalo/CSE-464-2026-ttrimalo/commit/6bbf9499873ee276e171679a15e6282161606a15>

Refactor 5:

<https://github.com/ttrimalo/CSE-464-2026-ttrimalo/commit/6ec93a8e5540123bf5139ee52d335c912fcdc4ca>

Applied strategy pattern for BFS and DFS:

<https://github.com/ttrimalo/CSE-464-2026-ttrimalo/commit/13125288a391f478aeb6727b97eb905494e04396>

Applied RandomWalkSearch:

<https://github.com/ttrimalo/CSE-464-2026-ttrimalo/commit/fcdf86f277581b6b0462df8946cbb34da4cbc692>